

DataCo SMART Supply Chain Analytics

Business Performance, Customer, Profitability & Delivery Analysis

- Disha Chaudhari
- Tools: **Python | PostgreSQL | Power BI**
- Dataset: **DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS**

1. Executive Summary

This project analyzes the DataCo SMART Supply Chain dataset to evaluate overall business performance, profitability, customer behavior, product performance, and delivery operations. The analysis combines Python-based exploratory data analysis, PostgreSQL analysis, and Power BI visualization to identify business risks and opportunities.

The dataset contains 180,519 order-item records representing 65,752 distinct orders and 20,652 customers. The dataset records approximately \$36.78 million in revenue and \$3.97 million in profit, resulting in an overall profit margin of approximately 10.78%.

The analysis identified delivery performance as a major operational concern. Approximately **57% of orders were classified as late**, with substantial differences across shipping modes. Second Class had an approximately **80% late-delivery rate** and an average delivery delay of about **2 days**, while First Class showed a 100% late-delivery rate in the dataset. In contrast, Standard Class had an average delivery delay close to zero. Late-delivery rates were also relatively consistent across markets, suggesting that the delivery issue is not isolated to a single geographic market.

Profitability analysis showed a clear association between discounting and lower margins. Profit margin decreased from approximately **13.09% for orders with no discount to 9.03% for the high-discount category**. This indicates that discounting should be evaluated alongside its impact on profitability rather than revenue alone.

Customer analysis showed that repeat and frequent customers are disproportionately valuable. Although these customers represented approximately **40.76% of customers, they generated about 78.25% of total revenue**. This indicates that customer retention and repeat purchasing are important contributors to business performance.

The analysis also identified strong revenue concentration across product categories. Fishing, Cleats, Camping & Hiking, and Cardio Equipment were among the highest-revenue categories. However, high revenue did not always correspond to the strongest profit margins, highlighting the importance of evaluating both revenue and profitability when prioritizing products and categories.

Overall, the analysis suggests three key priorities: **improve delivery reliability, manage discounting with greater attention to margin, and strengthen strategies for retaining high-value repeat customers**. These findings are presented through an interactive Power BI dashboard designed to support business-focused decision making.

2. Business Problem & Objectives

Business Problem

Supply-chain businesses must balance **revenue growth, profitability, customer value, and delivery reliability**. High sales alone do not necessarily indicate strong business performance if revenue is generated through excessive discounting, low-margin products, or operationally inefficient deliveries.

This project analyzes the DataCo SMART Supply Chain dataset to identify the major factors affecting commercial and operational performance and translate the findings into actionable business recommendations.

The analysis focuses on four core business areas:

- **Business performance:** revenue, profit, profit margin, and market contribution
- **Customer and product performance:** customer segments, purchasing behavior, categories, and products
- **Profitability:** discounting, order size, departments, and market-level profitability
- **Delivery operations:** shipping modes, delivery delays, and late-delivery patterns

Business Objectives

The analysis aims to answer the following questions:

1. **How is the business performing overall in terms of revenue, profit, and margin?**
2. **Which markets, customer segments, and product categories contribute most to revenue and profit?**
3. **How does discounting relate to profitability?**
4. **Which customer groups generate the greatest share of revenue?**
5. **Which shipping modes experience the greatest delivery problems?**
6. **Is late delivery concentrated in particular markets or widespread across the business?**
7. **What actions could improve profitability, customer value, and delivery performance?**

Analytical Approach

The project follows a business-driven analytical workflow:

Business Questions → Data Preparation → Feature Engineering → Exploratory Data Analysis → SQL Validation → Power BI Dashboard → Business Recommendations

3. Key Business Findings

3.1 Overall Business Performance

The business generated approximately **\$36.78 million in revenue** and **\$3.97 million in profit**, resulting in an overall profit margin of approximately **10.78%** across **65,752 orders** and **20,652 customers**.

The average order value was approximately **\$559.45**, while the average delivery delay was approximately **1 day**. However, the most significant operational concern was delivery reliability: approximately **57.33% of orders were classified as late**.

These results indicate that the business is generating substantial revenue, but operational performance—particularly delivery reliability—represents a significant area for improvement.

3.2 Revenue Performance

Revenue was relatively stable for most of the dataset, with monthly revenue generally around \$1 million. However, revenue declined sharply toward the end of the available period:

- **October 2017:** ~\$1.074M
- **November 2017:** ~\$0.627M
- **December 2017:** ~\$0.504M
- **January 2018:** ~\$0.332M

This decline should not automatically be interpreted as a genuine decline in business demand or performance. Order coverage and volume also change substantially toward the end of the dataset, suggesting that the decline may partly reflect a data-coverage or structural issue.

Therefore, further investigation using a complete and consistently sampled dataset would be required before making a business decision based on this trend.

3.3 Market Performance

Europe and LATAM were the two largest revenue-generating markets:

Market	Revenue Contribution
Europe	29.56%
LATAM	27.94%
Pacific Asia	22.49%
USCA	13.77%
Africa	6.24%

Together, **Europe and LATAM contributed approximately 57.50% of total revenue**, while Europe, LATAM, and Pacific Asia together contributed approximately **80%**.

Despite differences in revenue scale, market-level profit margins were relatively similar, ranging from approximately **10.37% to 11.14%**. This suggests that the primary difference between markets is their **scale of business**, rather than large differences in profitability.

3.4 Profitability and Discounting

Overall profit was approximately **\$3.97 million**, with an overall margin of **10.78%**.

A clear relationship was observed between discount level and profit margin:

Discount Category	Profit Margin
No Discount	13.09%
Low	11.48%
Medium	10.16%
High	9.03%

Profit margin decreased as discount intensity increased.

This represents an **association rather than proof of causation**. Other factors, such as product mix and order characteristics, may also influence profitability. Nevertheless, the pattern indicates that discounting should be evaluated based on **incremental revenue and margin impact**, rather than revenue growth alone.

3.5 Customer Value

Customer frequency analysis revealed a strong difference between customer count and revenue contribution.

Customer Group	Customers	Revenue Share
One-time	43.02%	7.60%
Occasional	16.23%	14.15%
Repeat	29.85%	48.87%
Frequent	10.91%	29.38%

Repeat and frequent customers represented approximately **40.76% of customers but generated approximately 78.25% of revenue.**

In contrast, one-time customers represented approximately **43.02% of customers but contributed only 7.60% of revenue.**

This indicates that repeat purchasing is a major contributor to revenue generation, making customer retention an important commercial priority.

3.6 Product and Category Performance

The dataset contains **118 products across 50 categories.**

The highest-revenue categories included:

- Fishing — ~\$6.93M
- Cleats — ~\$4.43M
- Camping & Hiking — ~\$4.12M
- Cardio Equipment — ~\$3.69M
- Women's Apparel — ~\$3.15M
- Water Sports — ~\$3.11M

However, high revenue did not always correspond to the highest profit margins. Several high-revenue categories, including **Camping & Hiking, Cardio Equipment, Water Sports, Shop By Sport, and Computers**, showed profit margins below the overall business margin.

This indicates that category prioritization should consider **both revenue contribution and profitability**, rather than ranking categories by sales alone.

3.7 Delivery and Operational Performance

Delivery reliability is the strongest operational issue identified in the analysis.

Overall delivery outcomes were:

- **Late:** 103,400

- **Early:** 43,366
- **On Time:** 33,753

Shipping mode performance varied substantially:

Shipping Mode	Late Delivery Rate	Average Delay
First Class	100%	~1 day
Second Class	~80%	~2 days
Same Day	~48%	~0.5 day
Standard Class	~40%	~0 days

The 100% late-delivery rate reflects the observed records in this dataset and should not be generalized beyond the dataset.

Second Class therefore represents a particularly important operational area because it combines a **high late-delivery rate with the highest average delay** among the major shipping modes.

At the market level, late-delivery rates were all close to **57%**, indicating that the delivery problem is relatively widespread rather than concentrated in a single geographic market.

3.8 Customer Concentration

The business does not appear to be dependent on a small number of individual customers.

The **top 10 customers contributed only approximately 0.25% of total revenue**, while the top individual customer contributed approximately **0.0286%**.

However, the broader customer-frequency analysis shows that repeat and frequent customers collectively generate a large share of revenue.

This distinction is important: **the business has strong dependence on repeat purchasing, but not on a handful of individual customers.**

4. Business Recommendations

4.1 Prioritize Delivery Reliability for Second Class Shipping

Evidence: Second Class shipments had an approximately **80% late-delivery rate** and an average delay of approximately **2 days**, making it the most significant shipping-mode performance issue in the analysis.

Recommendation:

Management should investigate the operational causes of Second Class delays, including carrier performance, scheduling assumptions, fulfillment processing time, and transportation capacity. Service-level targets should be reviewed to determine whether the current scheduled delivery times are realistic.

Expected business impact:

Improving Second Class reliability could reduce late deliveries, improve customer experience, and reduce operational disruptions.

4.2 Review Discounting Strategy Based on Profitability

Evidence: Profit margin declined from approximately **13.09% for orders without discounts to 9.03% for the high-discount category.**

Recommendation:

Discounting should be evaluated based on **incremental profit**, not revenue alone. High discounts should be targeted toward situations where the additional sales volume justifies the reduction in margin.

Management could consider establishing minimum margin thresholds for promotional campaigns and reviewing high-discount categories individually.

Expected business impact:

A more disciplined discounting strategy could help evaluate whether the additional revenue generated by promotions justifies the associated reduction in margin.

4.3 Focus Customer Retention Efforts on Repeat and Frequent Customers

Evidence: Repeat and frequent customers represented approximately **40.76% of customers but generated 78.25% of revenue.**

Recommendation:

Customer retention initiatives should prioritize converting occasional and one-time customers into repeat purchasers while maintaining relationships with existing frequent customers.

Potential strategies include personalized offers, post-purchase engagement, product recommendations, and targeted retention campaigns.

Expected business impact:

Increasing repeat purchase frequency could generate additional revenue without relying entirely on acquiring new customers.

4.4 Evaluate Categories Using Both Revenue and Margin

Evidence: Several high-revenue categories, including **Camping & Hiking, Cardio Equipment, Water Sports, Shop By Sport, and Computers**, showed profit margins below the overall business margin.

Recommendation:

Category performance should be evaluated using a combination of **revenue, profit, and profit margin**. High-revenue but lower-margin categories should be investigated for pricing, discounting, product costs, and product mix.

Expected business impact:

This approach can help management distinguish between categories that are simply high-volume and those that create strong economic value.

4.5 Investigate the Data Coverage Issue Before Acting on the Late Revenue Decline

Evidence: Revenue fell sharply from approximately **\$1.07M in October 2017 to \$0.33M in January 2018**, while order coverage also changes substantially during the same period.

Recommendation:

Before interpreting the decline as a genuine demand or business-performance problem, management should validate whether the underlying dataset has complete and consistent order coverage for the final months.

Expected business impact:

This prevents decisions from being made based on a potential data-quality or coverage issue.

4.6 Monitor Delivery Performance Across Markets, but Avoid Geographic Overreaction

Evidence: Late-delivery rates were approximately **56.6%–57.6% across markets**, showing very little variation.

Recommendation:

The company should treat late delivery primarily as a **system-wide operational issue**, rather than directing major corrective efforts toward one particular market.

Operational investigations should focus first on shipping modes and fulfillment processes, where the analysis shows much larger differences.

Expected business impact:

Resources can be directed toward the operational factors with the strongest evidence of performance differences instead of assuming geography is the primary cause.

5. Power BI Dashboard

An interactive Power BI dashboard was developed to communicate the key findings identified through Python and PostgreSQL analysis. The dashboard is structured into four pages, with each page focusing on a distinct business area.

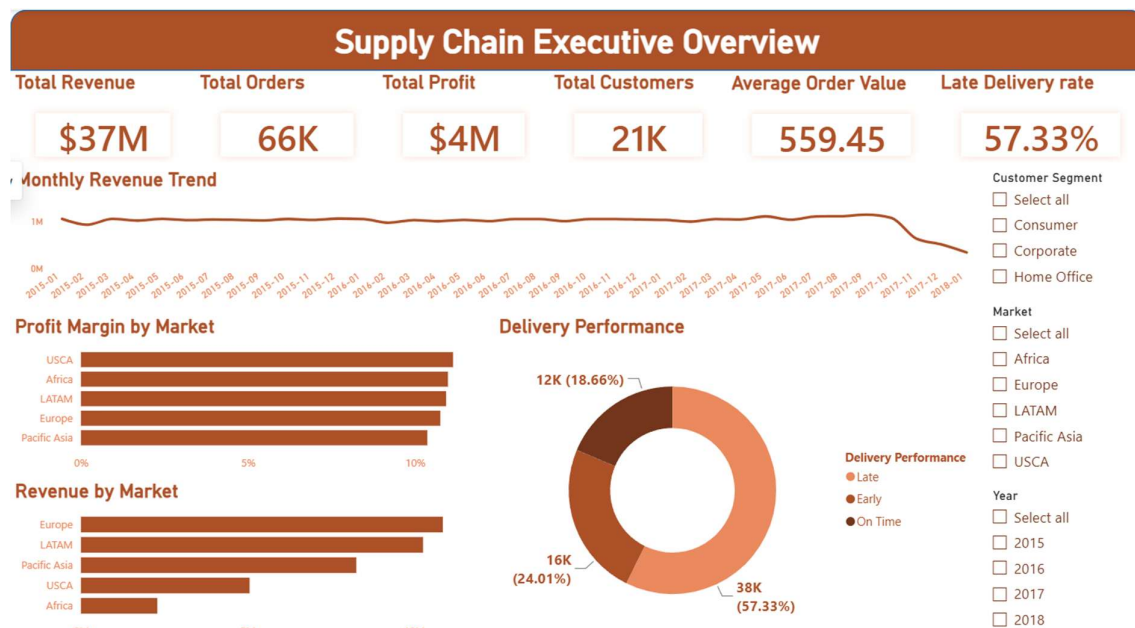
5.1 Supply Chain Executive Overview

The executive dashboard provides a high-level view of business performance through key performance indicators and summary visualizations.

It includes:

- Total Revenue
- Total Profit
- Total Orders
- Total Customers
- Total Quantity Sold
- Average Order Value
- Revenue trend over time
- Revenue and profit by market
- Overall delivery performance

This page is designed to allow management to quickly assess the overall health of the business before drilling into specific areas.



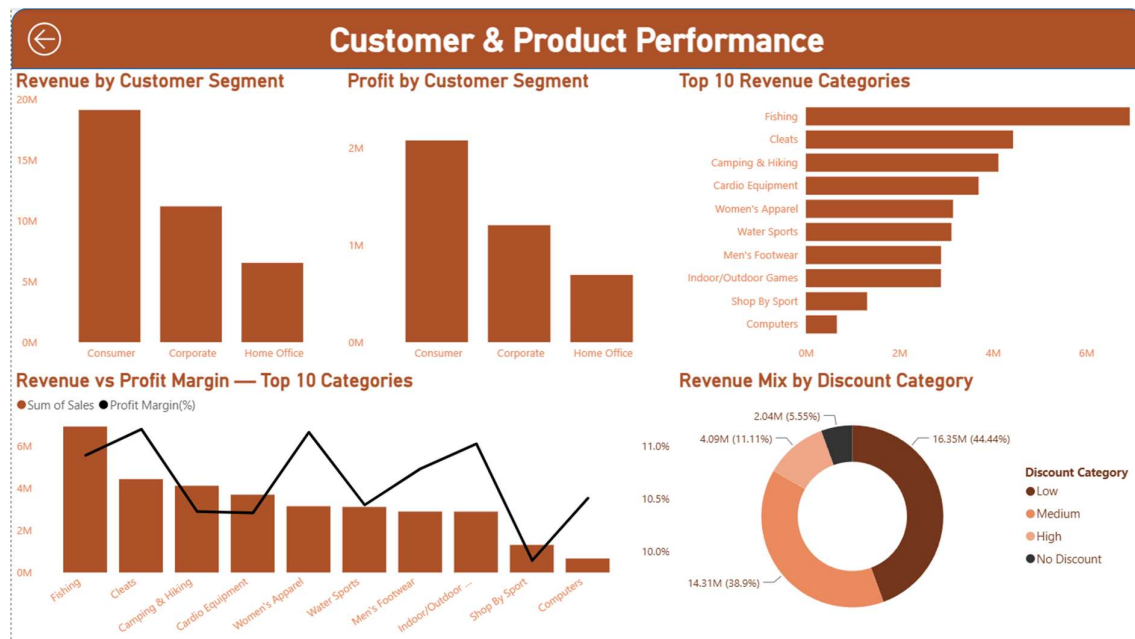
5.2 Customer & Product Performance

This page focuses on customer segments, product categories, and commercial performance.

Key analyses include:

- Revenue by customer segment
- Profit by customer segment
- Top 10 revenue-generating categories
- Revenue versus profit margin across major categories
- Revenue distribution by discount category

The page helps identify where revenue is being generated and whether high-revenue areas are also producing healthy margins.



5.3 Delivery & Operations

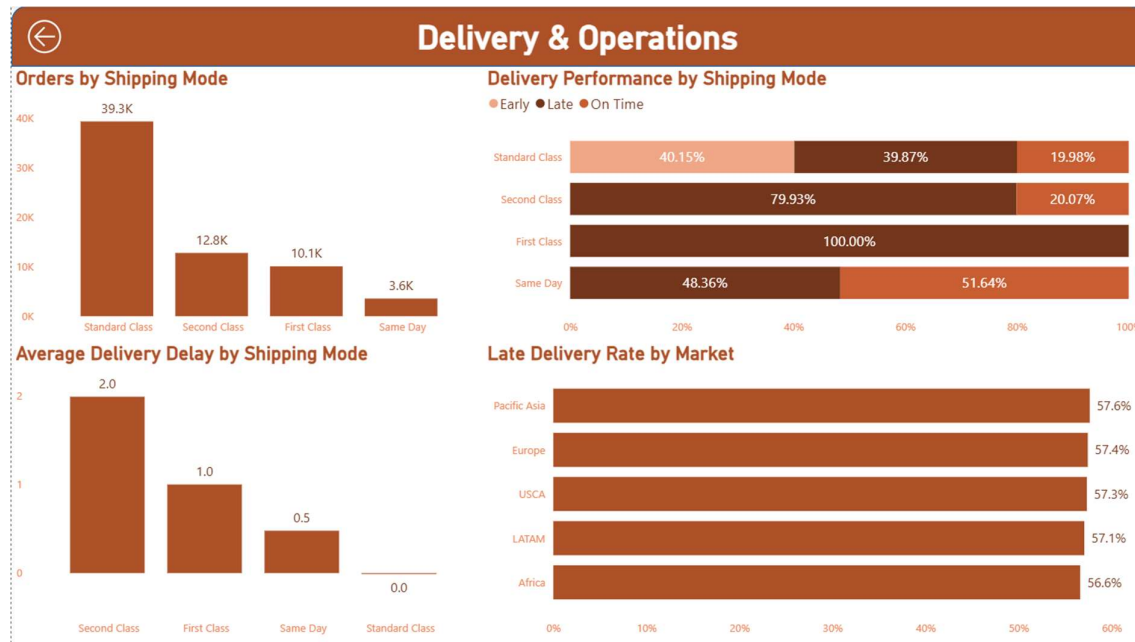
This page focuses on supply-chain execution and delivery reliability.

Key analyses include:

- Order volume by shipping mode
- Delivery performance by shipping mode
- Average delivery delay by shipping mode

- Late delivery rate by market

The dashboard highlights the significant variation in delivery performance between shipping modes and supports the recommendation to investigate Second Class shipping operations.



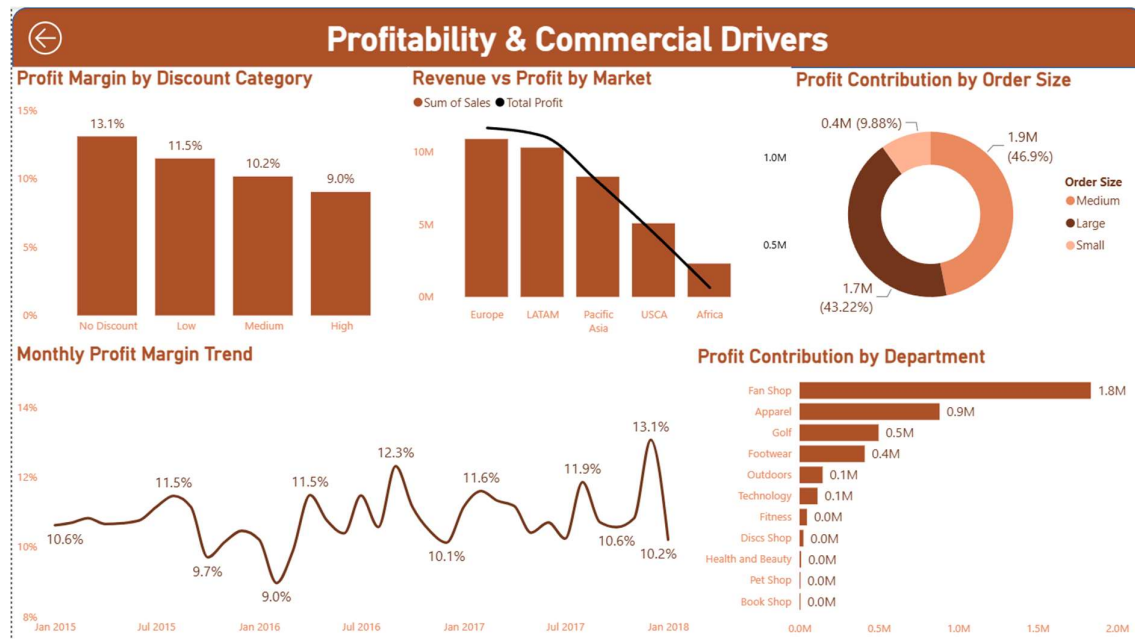
5.4 Profitability & Commercial Drivers

The final dashboard page focuses on factors associated with profitability.

It includes:

- Profit margin by discount category
- Revenue versus profit by market
- Profit contribution by order size
- Monthly profit margin trend
- Profit contribution by department

This page helps management evaluate the relationship between commercial scale, discounting, profitability, and operational business units.



6. Data Limitations

The analysis provides useful business insights, but several limitations should be considered when interpreting the results.

6.1 Historical Dataset

The analysis is based on a historical DataCo supply-chain dataset rather than live operational data. The findings therefore describe patterns within the available historical period and should not automatically be assumed to represent current business conditions.

6.2 Late-Period Revenue Coverage

Revenue shows a substantial decline toward the end of the dataset, particularly from October 2017 through January 2018. However, order volume and data coverage also change during this period.

Therefore, the decline should not automatically be interpreted as a genuine decline in business demand or performance without first validating that the final months contain complete and comparable data.

6.3 Observational Relationships

The analysis identifies associations but does not establish causality.

For example, profit margins decrease as discount levels increase. This indicates an association between discounting and lower margins, but it does not prove that discounts alone caused the reduction. Product mix, pricing, costs, and other factors may also contribute.

6.4 Order-Item Level Data

The source dataset contains order-item records rather than one row per order. As a result, analyses involving orders and customers must account for the underlying data grain.

Where appropriate, the analysis uses **distinct Order IDs** rather than simply counting rows to avoid overstating the number of orders.

6.5 Delivery Metrics

Delivery performance can produce different results depending on whether calculations are performed at the order-item level or distinct-order level. The Power BI dashboard uses distinct orders where appropriate for order-level metrics.

6.6 No Predictive Modeling

This project focuses on **descriptive and diagnostic analytics** rather than predictive modeling. The analysis identifies historical patterns and potential business actions but does not predict future sales, delivery risk, or customer behavior.

7. Conclusion

The DataCo SMART Supply Chain analysis provides a comprehensive view of the business across **financial performance, customers, products, profitability, and delivery operations**.

The dataset records approximately **\$36.78 million in revenue** and **\$3.97 million in profit**, with an overall profit margin of approximately **10.78%**. While revenue generation is substantial, the analysis identified several areas requiring management attention.

The most significant operational issue is **delivery reliability**, with approximately **57% of orders classified as late**. The particularly high late-delivery rates observed for Second Class and First Class shipping indicate that shipping-mode performance should be investigated.

Profitability analysis also identified a clear association between **higher discount levels and lower profit margins**. At the same time, customer analysis showed that **repeat and frequent customers generate a disproportionately large share of revenue**, making customer retention an important commercial priority.

The analysis also demonstrates that high revenue does not always correspond to high profitability. Evaluating revenue alongside profit and margin provides a more complete basis for prioritizing markets, categories, and business units.

Overall, the project demonstrates a complete **business-focused data analytics workflow**, from data preparation and feature engineering through exploratory analysis, SQL validation, dashboard development, and business recommendations. The resulting Power BI dashboard provides an interactive way for stakeholders to monitor performance and investigate the areas identified through the analysis.